

A Systematic Review for Crypto Currency Price Prediction using Machine Learning

*Note: Sub-titles are not captured in Xplore and should not be used

Pankaj
Computer Science and Engineering
(GJU S&T)
Guru Jambheshwar University of
Science and Technology
Hisar, India
pankajbishnoi1009@gmail.com

Abstract— The present research is reviewing the requirement of a novel approach in the case of cryptocurrency price prediction. Several types of research in the area of cryptocurrency and price prediction have been considered. Moreover, researches that are related to machine learning and deep learning are considered. The research paper considers the role of Artificial intelligence and machine learning in the price prediction of cryptocurrency. The issues in the case of previous research and the need for research in the area of cryptocurrency price prediction have been considered. The future scope of BTC, Wave, Ethereum, and Matic cryptocurrency value prediction is also discussed.

Keywords—Crypto currency, Deep Learning, Optimization, Performance, Prediction, Accuracy, recall value, Precision

I. INTRODUCTION

Crypto currency, a kind of digital cash, is protected against counterfeiting and double-spending by its built-in encryption. Decentralized networks based on block chain technology, which are widespread in various crypto currencies, are built on distributed ledgers enforced by an enormous network of computers. Someone (or a group of people) using the pseudonym "Satoshi Nakamoto" created Bit coin in 2009. Efforts to assist people manage their own money without relying on businesses or a government was launched in 2008 after the global financial crisis.

A. Cryptocurrency

Without the participation of a government or bank, this digital currency may be traded through a computer network.

These records are stored in digital ledgers, which are databases that use strong encryption to save transaction records and oversee currency production and verification. Crypto currency is a misnomer, as it does not refer to conventional money in the traditional sense. Instead, they are seen as a distinct class of assets that may be categorized as a commodity, security, or even a medium of exchange in and of themselves. In certain crypto systems, a group of individuals known as "validators" maintains the currency. In a proof-of-stake system, owners utilise their tokens as a kind of collateral. In exchange for a portion of their investment, they get ownership of the underlying token. Stackers of tokens are often compensated by network fees, newly minted tokens, or other means.

Cryptocurrency, unlike conventional money, does not exist in any tangible manner. Instead than being managed by a central bank, cryptocurrency is governed by a decentralized network (CBDC). Centralized crypto currencies are those that have been established or generated and then issued by a single

company. To serve as a public database for financial transactions, each cryptocurrency uses distributed ledger technology, often a blockchain, when it is deployed with decentralized governance.

This kind of digital asset or cash is referred to as "cryptocurrency." The use of encryption to safeguard and authenticate transactions is referred to as "cryptocurrency." There are already more than a thousand different types of crypto currencies. Bitcoin, the first decentralized cryptocurrency, was released as open-source software in 2009 under the name of the Bitcoin Core project. Since bitcoin's inception, a slew of other crypto currencies have emerged.

B. Role of AI in Crypto currency

Taking into account a stock's 52-week low/high and current market value while selecting a share script is meant by the artificial intelligence. In an effort to improve the accuracy of valuing a company's share capital, artificial intelligence (AI) optimization algorithms have been created.

C. Deep Learning

Deep learning is one of several subcategories of machine learning. Knowledge-rich output can only be generated by well-structured and controlled data. Deep neural learning is the term used to describe this kind of learning. The term "deep neural network" was coined by a single researcher. Algorithms are at the heart of machine learning. A large amount of data is taken into consideration while doing deep learning. Automated learning systems must be continuously supervised, according to new research. The programmer is responsible for defining all programming instructions. For example, a person trying to find a picture of a dog will have to provide the computer with a lot of information. When a search engine is self-aware and has access to a big store of information, the results are better. Because it is both quicker and more accurate, deep learning outperforms basic machine learning.

It's a well-known fact that education may be very effectively orchestrated. This behavior is known as hierarchical learning. In the wider environment, machine learning methods are getting more attention. These methods rely heavily on artificial neural networks.

D. Machine Learning

It's no secret that machine learning often makes use of artificial intelligence. When discussing artificial intelligence, the term "AI," which stands for "Artificial Intelligence," is often used. This is where the idea of clustering was first presented. This tool may be useful in circumstances involving reinforcement learning.

Machine learning is one of the most important uses of AI. Because of it, the system may develop and improve. The fact that it has grown suggests that it hasn't been programmed to its fullest potential. One-person computer programmes are the primary focus of machine learning.

Human teachers are no longer necessary since computers can now study on their own, thanks to advances in artificial intelligence. These scenarios do not need human intervention. The following are some of the benefits of machine learning:

1. As well as the financial sector and the healthcare sector, it is also widely used.
2. Devices have the ability to reduce their time cycles, resulting in increased resource consumption.
3. In the world of social media, this strategy is often used by firms such as Google and Face book.
4. Large and complicated process systems may benefit from machine learning.

E. Need and Motivation

An accurate prediction of Bitcoin prices requires a scalable and adaptable methodology. For this investigation, researchers will consult previous work in the fields of deep learning and machine learning. Researchers want to investigate the drawbacks of traditional deep learning systems, such as their inaccuracy and high time demands. It is essential that researchers first utilise optimization techniques to build efficient processes before comparing their results to those of previous investigations.

II. LITERATURE REVIEW

In 2013, A. Cheung [1] gained a great deal of media attention, he was mostly ignored by academics. This knowledge gap is filled as a consequence of their efforts. They employ a recently developed approach that is fairly excellent at recognising bubbles as part of our investigation on the prevalence of bitcoin bubbles. Between 2010 and 2013, there have been a number of short-lived bubbles, but the three major bubbles, each lasting between 66 and 106 days and causing the demise of Mt Gox, was the most famous.

For the first time, we're evaluating whether or not Bitcoin can serve as a supplement to other emerging market currencies. To be more precise, I examine how Bitcoin's value and volatility compare to those of other emerging market currencies. In this study, researchers found that there are ways to lessen Bitcoin's risks.

It was utilised by J. Liu [3] to investigate the relationship between volatility and returns of a popular cryptocurrency, as well as to research spill over from the cryptocurrency market to other financial markets. GARCH-in-mean model There is a statistically substantial transmission of shocks and volatility among the most popular digital currency tokens. In the United States and other large nations, the bitcoin market has statistically significant spillover effects.

Jumps have been detected in all three of the return series of 12 crypto currencies, according to E. Bouri in Inn 2020 [4]. It examines if the returns of various cryptocurrencies jump together, and the outcomes of their analyses show that, on the overall, they do (Ripple and Bytecoin). The presence of a cryptocurrency surge, it seems, increases the chance of an increase in other crypto currencies as well, according to these data. On the other side, co-jumping was a leap in trade volume

done in tandem. Trading volume increases are critical for the development of price rises in crypto currencies, as this recent research verifies prior findings on the influence of trading volume on volatility.

In 2020, N. Akbulaev [5] was analysing how bitcoin and Ethereum theoretically and practically interact with one other. Current developments in this field are examined by looking at literature on the underlying aspects and relationships between the prices of Bitcoin and Ethereum. In this article, author investigated the relationship between Ethereum and bitcoin prices, based on the concerns and research that are accessible. Using a mathematical model of cryptocurrency link analysis, it can be proven that the relationship was fairly tight. To limit risk while trading on exchanges like Gemini, it was possible to exploit the correlation between Bitcoin and Ethereum's prices to diversify and mitigate risk.

A 2018 research by Baur D. [6] examined whether Bitcoin is a medium of exchange or an asset, as well as how it is now utilised and how it will be used in the future, given its characteristics. Bitcoin's statistical characteristics were distinct from those of traditional asset classes like equities, bonds, and commodities in both calm financial times and stormy periods. Data from Bitcoin accounts shows that rather than being used as a replacement currency or a medium of exchange, the money was mostly used for speculating purposes.

[7] Corbet analysed the temporal and frequency relationships between leading cryptocurrencies and a wide range of financial assets in 2018. According to the findings of this research, these assets are distinct from financial and economic assets in terms of their structure. Cryptocurrency investments may provide diversification advantages for short-term investors, according to research. Economic and financial shocks have a long-term effect on relationships.

E. Turkedjiev [8] used the ANN in 2013 to forecast short-term stock values, notably in the banking industry. Because of its high nonlinearity, artificial neural networks may be utilized to analyze financial share time-series (ANN). There was a comparison in this study between the ANN and a more traditional linear regression model. It was possible to identify nonlinearity by comparing the experimental results from ANN and linear regression models.

Also in 2007, two new factors appeared that had a significant impact on the price discrepancies between the A and H shares in 2010. The price gap between A and H equities may be narrowed with the help of certain policy ideas.

Bank share prices were affected by a variety of important factors such as after-tax profits per share, GDP, and the market index according to L. Guoyi [10]. An related test model evaluates and tests this data in order to establish whether investors may use it as a reference for making investment decisions if there is a relationship between it and bank share prices at the end of the reporting period

According to S.Beng Ho [11], a successful general learning machine must be "general" in the sense that it can learn to solve many various sorts of problems, but it must also learn to solve them quickly and continue to learn rapidly in a non-stationary environment. A new research strategy is being recommended for the development of truly ubiquitous and adaptable intelligent computers that can reason for themselves. We examined the existing limits of popular

learning techniques, such as reinforcement learning, and offered new methodologies to address these gaps in the current learning process.

D. Zhang [12] offered an overview of the three strategies mentioned and their possible use in smart grids in 2018.

TABLE I. LITERATURE SURVEY

S No.	Author / Year	Title	Methodology	Conclusion
1	Cheung A. / 2015	Application of Phillips-Shi-Yu (2013) approach on bitcoin prices from Mt. Gox: Crypto-currency bubbles	Prediction Mechanism	Research work did prediction for crypto return
2.	Carrick J. / 2016	Emerging market currencies may benefit from the addition of Bitcoin.	Pattern analysis	Volatile patterns were analyzed.
3	Liu J. & Serletis A. / 2019	The bitcoin market is experiencing significant volatility.	Prediction Mechanism	Various factors such as inflation and prices were considered during prediction.
4	Bouri E. / 2019	Is the price of Bitcoin and other cryptocurrencies correlated?	Prediction Mechanism	More efficiency has been introduced for crypto prediction.
5	Akbulaev N. / 2020	Ethereum's Price Relationship with Volume and Bitcoin's Price: Correlation and Regression Analysis	Inter temporal relation analysis	Returns and risk factors considered to establish intertemporal relation
6	Baur D. / 2018	Speculative assets or a means of exchange: what is the role of Bitcoin?	Correlation analysis	Influencing factors for bank share price were considered.
7	Corbet S. / 2018	Discovering how cryptocurrency and other financial assets interact dynamically	Empirical analysis	Impact factors for share prices were considered
8	Alagidede / 2012	Inflation and stock returns: quantified regression evidence	Regression technique	Research provided more accuracy
9	E. Turkedjiev / 2013	Share Price Short-Term Trading in the UK Banking Sector Using an Artificial Neural Network Model	ANN	ANN made the share price prediction smarter.
10	W. Kang / 2013	Oil shocks, policy uncertainty, and the stock market's comeback	Cost prediction	Uncertainties have been considered during price prediction
11	S. Kim / 2014	Internet message posts and the predictability of stock returns reveal investor emotion.	Prediction Mechanism	Work found suitable for real life
12	T. Hendershott / 2014	Predictability of stock returns and liquidity provision	Prediction Mechanism	Liquidity provision has been considered
13	X. Li / 2014	Improved intra-day stock return forecasting by including both market news and stock price information	Quantitative Prediction	Research has enhanced quantitative stock return prediction.
14	S.Beng Ho / 2016	AI's future depends on its ability to think deeply and quickly.	Deep learning	Research proposed quick learning and AI based solution.
15	Tang / 2016	In software defined networking, a network intrusion detection system employs a deep learning technique.	Deep learning	Research is playing significant role in security by providing intrusion detection.
16	R. McKenna / 2016	High-end cluster runtime and IO traffic estimates using machine learning	Machine learning	Research provided high end cluster to support prediction
17	Sunpreet Kaur / 2016	Algorithms for Machine Learning	Machine learning	Several machine learning system have been analyzed
18	S. Wawre / 2016	Methods for Machine Learning-Based Sentiment Analysis	Machine learning	There is need to increase the training performance during machine learning operation.
19	Yin / 2017	Recurrent Neural Networks (RNNs) may be used for intrusion detection.	Recurrent Neural Networks	RNN model provided efficient solution for IDS detection.
20	Sandra Sendra / 2017	Using software-defined networks to include artificial intelligence into a routing protocol	Artificial Intelligence	Research provided solution for routing protocol with support of SDN system.
21	D. Zhang / 2018	Deep Learning and Reinforcement Learning in Smart Grids: A Review of Research and Practice	Deep Learning and Reinforcement Learning	Reinforcement learning system has been reviewed.
22	F. Wang / 2018	In a roaming market, how to best price user-initiated data plan sharing?	Optimization	Optimized cost has been predicted.
23	X. Qiao / 2019	Optimal pricing strategy and the economic impact of product sharing based on B2C platform analysis	Optimization	Research is providing best solution for B2C
24	Arti / 2019	The Use of Machine Learning Techniques for Indian Premier League Opinion Mining	Opinion Mining	Accuracy has been focused during machine learning for opinion mining.
25	Roni Bhowmik / 2020	A Systematic Review of the Literature on Stock Market Volatility and Return	Volatility and Return Analysis	Research has reviewed analysis for volatility and return
26	Samiksha Mame / 2021	Predicting Price of Cryptocurrency – A Deep Learning Approach	Deep learning	Deep learning system has been reviewed.
27	Shen Xiaohong / 2021	Machine Learning-Based Investor Sentiment Mediates the Impact of Research Reports on Stock Returns.	Machine Learning	Investor sentiment played significant role during stock price prediction.
28	Patrick Jaquart/2021	Short-Term Bitcoin Market Prediction via Machine Learning	Machine Learning	Research work did prediction for Bitcoin

29	Sumit Ranjan / 2022	Bitcoin Price Prediction: A Machine Learning Sample Dimension Approach	Machine Learning	Machine learning is used to get the optimization value of crypto.
30	Khobragade / 2022	Forecasting of Cryptocurrency Values using Machine Learning	Machine Learning	Research has reviewed analysis for volatility and return
31	S. Hansun / 2022	A.Q.M. Multivariate cryptocurrency prediction: comparative analysis of three recurrent neural networks approaches	Recurrent neural networks	Recurrent neural networks system has been reviewed.

A. Time span of research

There have been many research [1–3] on forecasting bitcoin returns. The inter-temporal link between anticipated returns and risk has been studied [4] and efficient crypto return prediction tests have been conducted [5]. After that, the correlation of factors that affect the price of cryptos in the banking industry was explored [6]. An empirical research was then done to assess the impact of numerous factors on the price of cryptocurrencies. [7] [8] There were studies that made use of quantized regression as well as inflation. Researchers have developed a new ANN model [9] and a novel cost predication approach [10] to predict stock prices. Other research looked at investor sentiment and the availability of liquid assets to predict stock returns [11, 12]. Several research have focused on the incorporation of market news and stock price information to improve quantitative intra-day stock returns

[13]. Researchers are using deep learning [14, 15] and machine learning [16, 17] to enhance prediction processes. Machine learning techniques such as RNN [19] were employed to categorise the emotions of investors [18]. AI and deep learning research become more popular as a consequence [20, 21]. Studies on optimization and opinion mining were conducted thereafter. These researchers also considered volatility, return analysis, and reports on stock returns [22, 23, 24]. In order to predict price of cryptocurrency [26], several deep Learning Approach were introduced and Mediated effect of machine learning based investor sentiment [27] were used to predict stock returns. Further research were made for short term bitcoin market prediction [28, 29] using machine Learning. Such research played significant role in forecasting of Cryptocurrency Values [30]. Comparative analysis of recurrent neural networks approaches were made for Multivariate cryptocurrency prediction [31] afterward.

2002-2007	2008-2012	2013-2017	2018-2021
• Cryptocurrency • Prediction Mechanism [1] • Pattern analysis [2] • Inflation [3] • Return predictability [4]	• Inter temporal relation analysis [5] • Correlation analysis [6] • Empirical analysis [7] • Regression technique [8]	• ANN [9] • Cost prediction [10] • Quantitative prediction [13] • Deep learning [14,15,20,26] • Machine learning [16,17,18] • RNN [19]	• Reinforcement learning [21,31] • Optimization [22,23] • Opinion mining[24,30] • Market volatility [25] • Investor sentiment analysis sentiment analysis[27] • Bitcoin Price Prediction [28,29]

Fig. 1. Time line of research

III. COMPARISON OF FEATURES

Artificial intelligence, on the other hand, has made some strides forward. The current state of research raises questions about how little attention is being paid to practical challenges. Shareholder value maximization has not been a part of popular wisdom. So far, researchers have only looked at the price of a coin and the market volatility to arrive at their conclusions. Deep learning algorithms have taken a long time to develop and test. For the trustworthiness to rise, it is required to improve accuracy, recall, and f-score. Following table is presenting comparative analysis of features that were available and not available in previous researches.

TABLE II. FEATURE COMPARISON

Citation	Stock	Crypto-currency	Deep learning	Prediction	Multi-variant
[25]	Y	N	N	N	N
[26]	N	Y	Y	Y	N
[27]	Y	N	Y	N	N
[28]	N	Y	Y	Y	N
[29]	N	Y	Y	Y	N
[30]	N	Y	N	N	Y
[31]	N	Y	Y	Y	Y

IV. NEED OF RESEARCH

The development of a scalable and adaptive model for the prediction of cryptocurrency prices is urgently required. Prior Deep Learning and Machine Learning studies will be considered in this study. We want to conduct a study on the accuracy and computing cost of a conventional deep learning system. There should also be an evaluation that compares current methods with those that have been offered for forecasting future prices of well-known cryptocurrencies such as Bitcoin and Matic, Ethereum, Waves and BTT. There is need to propose research that should study current techniques and issues for crypto currency price predication in order to propose a novel approach for price prediction of BTC, Ethereum, Matic and Wave crypto currencies.

V. CONCLUSION

Current study examines the need for a new strategy to predicting the value of crypto currencies. Many studies in the field of digital money and forecasting its value have been explored. Additionally, research in the field of machine learning and deep learning is taken into account. In this study, the function of artificial intelligence and machine learning in predicting the price of cryptocurrencies is examined in detail. Problems of previous research and requirement of research for cryptocurrency price prediction

are considered. The future value projection of BTC, Wave, Ethereum, and Matic cryptocurrencies is also examined in this article. Cryptocurrency values may be predicted using a wide range of factors. Many fund managers and private investors depend on these kinds of research when making investing choices. To eliminate scripts that aren't appropriate for a certain situation, scientists have developed a system that is both scalable and adaptive.

REFERENCES

- [1] Cheung A., Roca E. & Su J. Crypto-currency bubbles: an application of the Phillips–Shi–Yu (2013) methodology on Mt. Gox bitcoin prices. *Applied Economics*. 47, 2348–2358 (2015).
- [2] Carrick J. Bitcoin as a complement to emerging market currencies. *Emerging Markets Finance And Trade*. 52, 2321–2334 (2016).
- [3] Liu J. & Serletis A. Volatility in the cryptocurrency market. *Open Economics Review*. 30, 779–811 (2019).
- [4] Bouri E., Roubaud D. & Shahzad S. Do Bitcoin and other cryptocurrencies jump together?. *The Quarterly Review Of Economics And Finance*. 76 pp. 396–409 (2020).
- [5] Akbulaev N., Mammadov I. & Hemdullayeva M. Correlation and Regression Analysis of the Relation between Ethereum Price and Both Its Volume and Bitcoin Price. *The Journal Of Structured Finance*. 26, 46–56 (2020).
- [6] Baur D., Hong K. & Lee A. Bitcoin: Medium of exchange or speculative assets?. *Journal Of International Financial Markets, Institutions And Money*. 54 pp. 177–189 (2018).
- [7] Corbet S., Meegan A., Larkin C., Lucey B. & Yarovaya L. Exploring the dynamic relationships between cryptocurrencies and other financial assets. *Economics Letters*. 165 pp. 28–34 (2018).
- [8] Alagidede, P., & Panagiotidis, T. (2012). Stock Returns and Inflation: Evidence from quantiled regressions. *Economic Letters*, 117, 283–286.
- [9] E. Turkedjiev, M. Angelova & K. Busawon, "Validation of Artificial Neural Network Model for Share Price UK Banking Sector Short-Term Trading," 2013 UKSim 15th International Conference on Computer Modelling & Simulation, Cambridge, 2013, pp. 6–11, doi: 10.1109/UKSim.2013.31.
- [10] Kang, W., & Ratti, R. A. (2013). Oil shocks, policy uncertainty and stock market return. *Journal of International Financial Markets, Institutions and Money*, 26, 305–318.
- [11] Kim, S., & Kim, D. (2014). Investor sentiment from internet message postings and the predictability of stock returns. *Journal of Economic Behavior and Organization*, 107, 708–729.
- [12] Hendershott, T., & Seasholes, M. S. (2014). Liquidity provision and stock return predictability. *Journal of Banking and Finance*, 45, 140–151.
- [13] Li, X., Huang, X., Deng, X., & Zhu S. (2014). Enhancing quantitative intra-day stock return prediction by integrating both market news and stock price information. *Neurocomputing*, 142, 228–238.
- [14] S.Beng Ho, "Deep Thinking and Quick Learning for Viable AI", Future Technologies Conference, Vol. 6, Issue 7, 2016.
- [15] Tang, Tuan A., LotfiMhamdi, Des McLernon, Syed Ali Raza Zaidi, and MounirGhogho. "Deep learning approach for network intrusion detection in software defined networking." In 2016 International Conference on Wireless Networks and Mobile Communications (WINCOM), pp. 258–263. IEEE, 2016.
- [16] R. McKenna, S. Herbein, A. Moody, T. Gamblin, and M. Taufer, "Machine learning predictions of runtime and IO traffic on high-end clusters," Proc. - IEEE Int. Conf. Clust. Comput. ICC3, pp. 255–258, 2016, doi: 10.1109/CLUSTER.2016.58.
- [17] Sunpreet Kaur, SonikaJindal, "A Survey on Machine Learning Algorithms", International Journal of Innovative Research in Advanced Engineering (IJIRAE), Vol. 3, Issue 11, 2016.
- [18] S. Wawre et. al. "Sentiment Classification using Machine Learning Techniques", International Journal of Science and Research (IJSR), Vol., 5 Issue 4, 2016.
- [19] Yin, Chuanlong, Yuefei Zhu, JinlongFei, and Xinzhen He. "A deep learning approach for intrusion detection using recurrent neural networks." *Ieee Access* 5 (2017): 21954–21961.
- [20] Sandra Sendra, Albert Rego, Jaime Lloret, Jose Miguel Jimenez, Oscar Romero, (2017), Including Artificial Intelligence in a Routing Protocol Using Software Defined Networks, ICC2017: WT04-5thIEEE International Workshop on Smart Communication Protocols and Algorithms (SCPA 2017).
- [21] D. Zhang, X. Han, and C. Deng, "Review on the Research and Practice of Deep Learning and Reinforcement Learning in Smart Grids," *CSEE Journal of Power and energy system*, vol. 4, Issue 3, 2018.
- [22] F. Wang, L. Duan & J. Niu, "Optimal Pricing of User-Initiated Data-Plan Sharing in a Roaming Market," in *IEEE Transactions on Wireless Communications*, vol. 17, no. 9, pp. 5929–5944, Sept. 2018, doi: 10.1109/TWC.2018.2851578.
- [23] X. Qiao, D. Shi & F. Xu, "Optimal pricing strategy & economic effect of product sharing based on analysis of B2C sharing platform," 2019 16th International Conference on Service Systems & Service Management (ICSSSM), Shenzhen, China, 2019, pp. 1–6, doi: 10.1109/ICSSSM.2019.8887720.
- [24] Arti, Kamanksha Prasad Dubey, Sanjay Agrawal, "An Opinion Mining For Indian Premier League Using Machine Learning Techniques" 2019 IEEE
- [25] Roni Bhowmik and Shouyang Wang. (2020). Stock Market Volatility and Return Analysis: A Systematic Literature Review. *Entropy* 2020, 22, 522; doi:10.3390/e22050522.
- [26] Samiksha Marne, Shweta Churi, Delisa Correia, Joanne Gomes, 2021, Predicting Price of Cryptocurrency – A Deep Learning Approach, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) NTASU – 2020 (Volume 09 – Issue 03).
- [27] XiaohongShen, Gaoshan Wang and Yue Wang. (2021). The Influence of Research Reports on Stock Returns: The Mediating Effect of Machine-Learning-Based Investor Sentiment. Volume 2021. Article ID 5049179. <https://doi.org/10.1155/2021/5049179>
- [28] Jaquart, Patrick & Dann, David & Weinhardt, Christof. (2021). Short-Term Bitcoin Market Prediction via Machine Learning. *The Journal of Finance and Data Science*. 7. 10.1016/j.jfds.2021.03.001.
- [29] Ranjan, Sumit & Kayal, Parthajit & Saraf, Malvika. (2022). Bitcoin Price Prediction: A Machine Learning Sample Dimension Approach. *Computational Economics*. 1–20. 10.1007/s10614-022-10262-6.
- [30] Khobragade, A. K., Keskar, O. C., Deshmukh, P. G., & Chopade, R. M. (2022). Forecasting of Cryptocurrency Values using. May. <https://www.ijraset.com/best-journal/forecasting-of-cryptocurrency-values-using-machine-learning>
- [31] Hansun, S., Wicaksana, A. & Khaliq, A.Q.M. Multivariate cryptocurrency prediction: comparative analysis of three recurrent neural networks approaches. *J Big Data* 9, 50 (2022). <https://doi.org/10.1186/s40537-022-00601-7>